SimpleFT8 — Two Antennas, One Comparison

What does Diversity actually deliver? — 40m FT8, first field results

Period: 2026-04-20 to 2026-04-24 · 15s cycles evaluated: 21881

Summary:

With Diversity Standard (blue) I decoded on average +91% more stations than with a single antenna.
Including 'rescued' stations, this rises to up to +127%.
Diversity DX (orange) delivers +126% — less than Standard, but targeted at weak DX signals.

Note on antenna setup: ANT1 is a Kelemen DP-201510 (fan dipole for 20m/15m/10m) — operated off-band on 40m and therefore suboptimal on this freq

What do the three modes mean?

Normal (grey): One antenna, no special logic — just like WSJT-X. This is the reference baseline.
Diversity Standard (blue): Two antennas. The system automatically picks the antenna with more stations.
Diversity DX (orange): Two antennas. Picks the antenna with the weakest DX signals (below −10 dB).
Rescue (green caps): Stations ANT1 couldn't hear — ANT2 decoded them anyway.

How was it measured?

Setup, time period and a bit of background

The Setup

Measured on 40m FT8 using a FlexRadio FLEX-8400M — two antenna ports, same frequency.
Period: 2026-04-20 to 2026-04-24. Data available across the full day (05–23 UTC), night measurements ongoing.
Cycles: Normal 6744 (4 day/s) | Diversity Standard 6398 (4 day/s) | Diversity DX 8739 (4 day/s).
Each FT8 cycle lasts 15 seconds — the app counts decoded stations per cycle.

Why not just use the daily average?

Good question — I did it that way at first too. But if one day had only 20 cycles and another had 500, the short day would carry the same weight. That's unfair. So all individual values are pooled and averaged directly — regardless of which day. This gives a picture much closer to reality.

What are 'rescued stations' (Rescue)?

Imagine a station transmitting so weakly that ANT1 receives it below −24 dB.
That is the FT8 decoding threshold — below that, decoding is practically impossible.
ANT2 receives the same station with slightly more signal and decodes it anyway.
We call this 'Rescue'. The green caps in the charts show exactly these stations.
Whether you count them or treat them separately — that's up to you. Both values are in this report.

Where does the raw data come from?

SimpleFT8 writes a small Markdown file per hour containing all cycle values.
No pre-computed averages — raw data only. The analysis script calculates everything fresh.
Want to check? statistics/ in the GitHub repo, all public.

Main Results — Comparison Table

40m FT8 · Pooled Mean

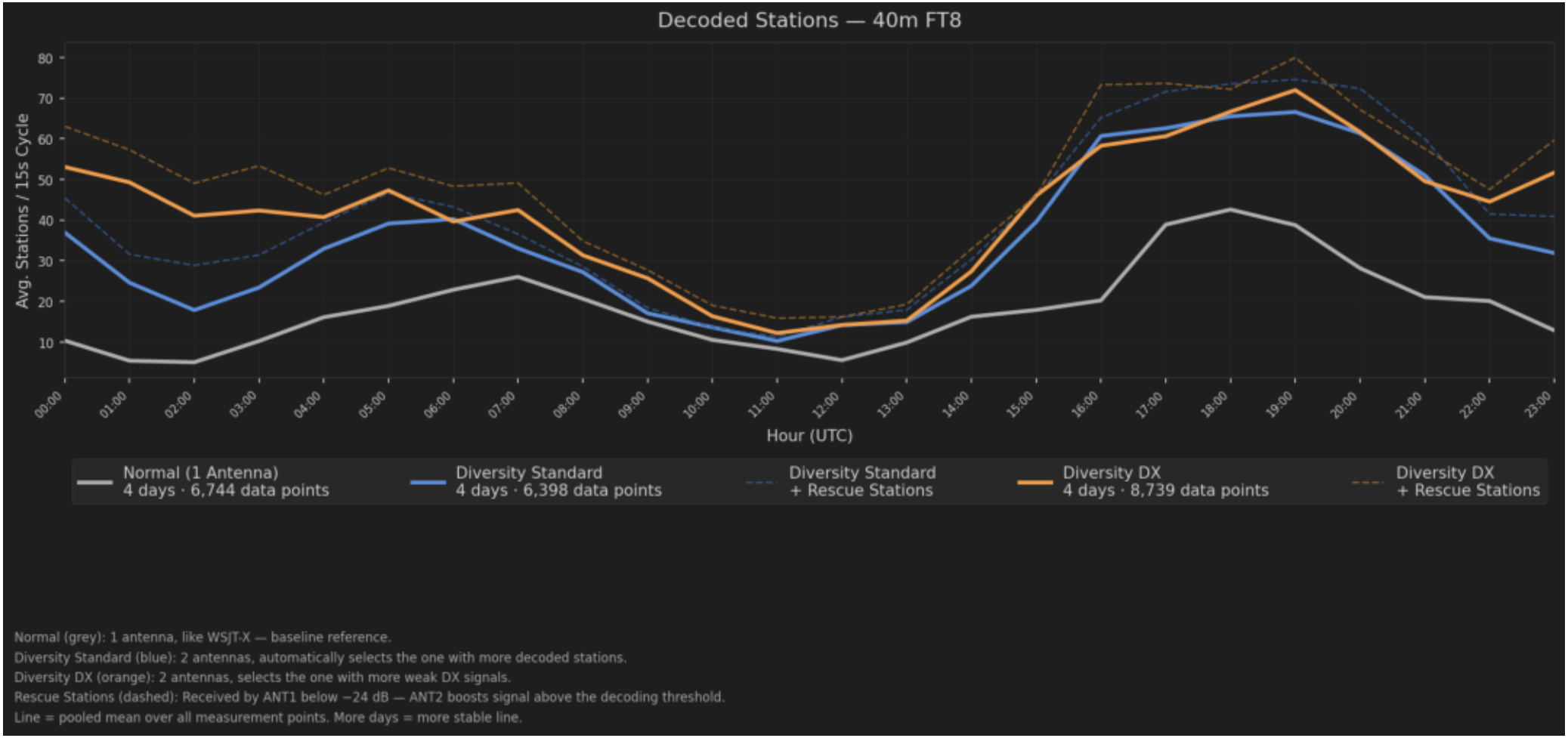
Mode	Avg Sta. /Cycle	vs Normal w/o Rescue	vs Normal + Rescue	Rescue only	Days	Cycles
Normal (1 Antenna	18.5	—	—	—	4	6744
Diversity Standard	35.3	**+91%**	**+127%**	**+36%**	4	6398
Diversity DX	41.7	**+126%**	**+160%**	**+34%**	4	8739

Avg Sta./Cycle = mean over all individual cycles directly (not first day averages). Rescue = ANT1 below −24 dB, ANT2 decoded anyway.

Note: ANT1 (Kelemen DP-201510) is operated off-band on 40m (design band: 20m/15m/10m) — ANT2 (house gutter ~15m) happens to be better suited for 40m. Gains represent

Stations per Hour — all three modes compared

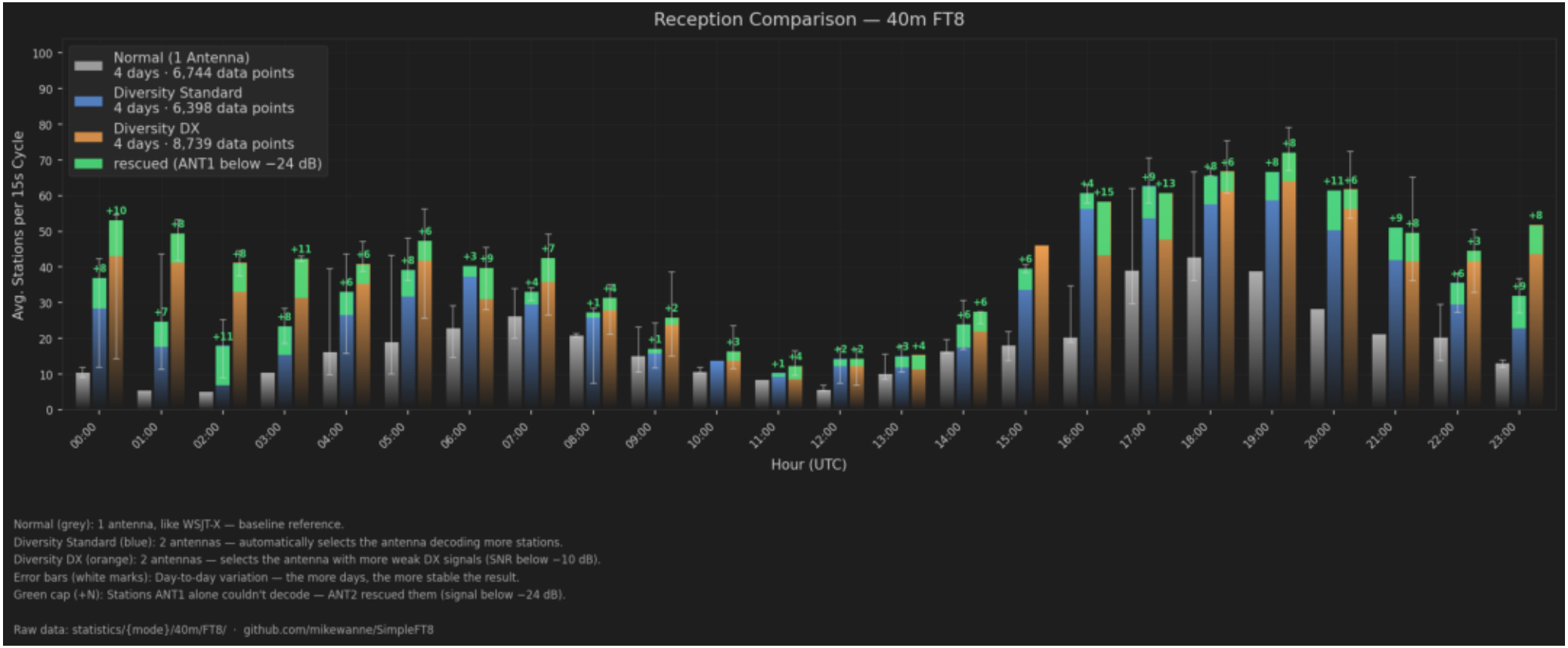
40m FT8 — line = mean, shaded band = day-to-day variation



You can clearly see how the grey line (Normal, one antenna) almost always falls below blue and orange. The shaded band shows day-to-day variation — as more days are added, the band will narrow.

Direct Comparison — bars per hour, three modes side by side

40m FT8 — Normal (grey) | Diversity Standard (blue) | Diversity DX (orange) | Rescue caps (green)



Green caps (+N) on top of bars show stations that ANT1 couldn't decode — signal below −24 dB. ANT2 rescued them anyway. Whether to count them or not — your call. White error bars show day-to-day variation.

The Green Caps — count them or not?

Rescue stations: What are they and what do they tell us?

What's this about?

In the diversity chart you can see small green caps with a +N on top of some bars.
These are stations ANT1 couldn't decode — their signal was below −24 dB, too weak.
ANT2 still received and decoded them. Over the measurement period this averaged
Ø 6.6 stations per hour for Diversity Standard, and Ø 6.3/h for Diversity DX.

Why should you count them in?

Because the QSO is real for the operator — regardless of whether ANT1 or ANT2 made it possible.
These stations would never have made it into the log with a single antenna.
This is not a measurement artifact — it's exactly the point of running a second antenna.
With Rescue: Standard +127%, DX +160% — that's the full picture.

Why might you leave them out?

SNR values fluctuate within the 15 seconds of a cycle — a station just
below −24 dB might be above that threshold in the next cycle.
The −24 dB threshold is an empirical value, not a law of nature.
For a clean comparison with other systems, counting only what ANT1 directly decoded is more conservative.

That's why this report always shows both numbers: without Rescue (the conservative value) and with Rescue (what you get in real operation). Ta

What can you take away from this?

Conclusions and what gets measured next

What is clearly visible:

Diversity Standard consistently delivers between 91% and 127% more stations across all measurement days — depending on whether you count rescued stations or not. This is not coincidence, it repeats.
Diversity DX sits at 126% without Rescue — less than Standard, but DX deliberately optimises for the weakest signals. If you do a lot of DX, that makes sense.
The strongest effect was at 01:00 UTC with +357% — typically the time when the band is just opening or closing.

What cannot be said yet:

Whether results transfer 1:1 to other antenna combinations — probably not.
ANT1 operates off-band on 40m, ANT2 happens to be better suited there.
This explains the particularly large gain — with two matched 40m antennas it would be smaller.
Contest operation, geomagnetic storms, poor conditions — not yet systematically tested.
Whether this works identically on other transceivers — tested only on the FLEX so far.

What comes next:

Night measurements on 40m are ongoing — 40m is often significantly better in the evening.
More days so the bars in the chart stabilize and variation decreases.
20m tests are planned — there ANT1 operates within its design band (20m/15m/10m) and the 20m band is generally easier to receive. Different
This report updates automatically whenever new data comes in.

Raw data: statistics/ in the repo · github.com/mikewanne/SimpleFT8 · DA1MHH